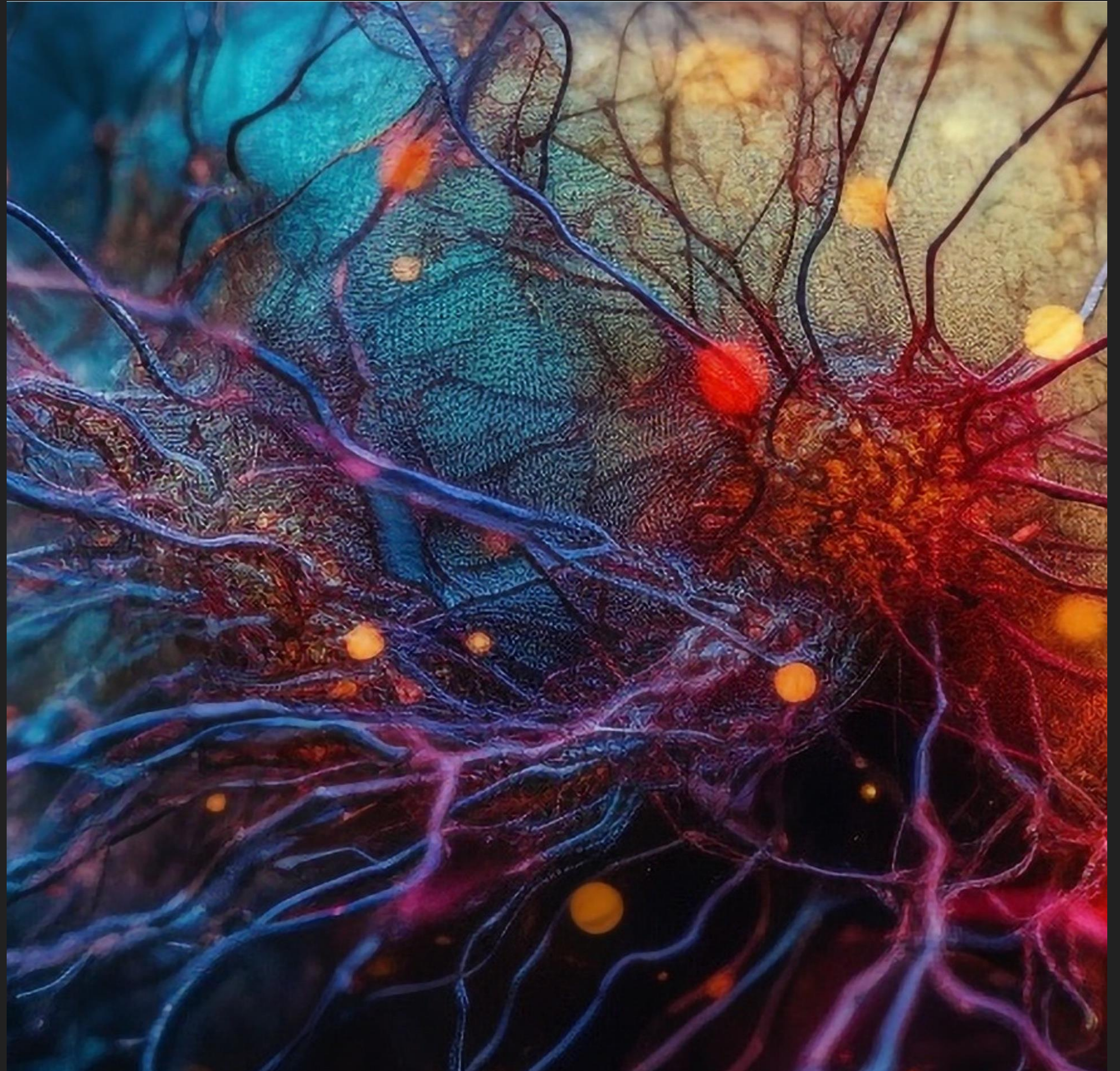


DNS Security: The Hidden Foundation of Cyber Resilience

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Agenda



- **Why DNS Matters**
- **DNS Threats Today**
- **AI-Driven Detection**
- **Building Resilience**

Why DNS Matters



Foundation of Internet

Every connection depends on DNS—emails, websites, cloud apps, everything.



Critical Infrastructure

Without DNS, your entire digital infrastructure stops working instantly.



Primary Attack Vector

Attackers exploit DNS for malware distribution, data theft, and service disruption.



DNS Threats Today

DNS is the #1 attack vector—here's what attackers are doing:

83%

of breaches involve DNS

45%

exploit DNS for C2 commands

67%

use DNS to exfiltrate data

AI-Powered Threat Detection

✗ The Problem:

- Manual detection misses 60% of attacks
- Rule-based systems create false positives
- Advanced threats hide in normal traffic
- Human analysis can't scale

✓ EfficientIP AI Engine:

- ML detects anomalies in real-time
- 95%+ accuracy on advanced attacks
- Zero-day threat detection
- Autonomous response & blocking

What Happens When DNS Fails



Ransomware Distribution

Attackers redirect users to malicious sites via DNS spoofing



Data Exfiltration

Sensitive data tunneled through DNS queries undetected



Service Disruption

DDoS attacks paralyze entire organizations instantly



Credential Theft

DNS hijacking captures credentials and authentication tokens



Three Pillars of DNS Security

**Granular
Visibility**



Detection

Real-time visibility
into all DNS traffic

**Real-Time
Analytics**



Prevention

Block malicious domains
and threats instantly

**Automation
& Orchestration**



Response

Automated incident
response & recovery

Building a Resilient DNS Foundation

- ✓ Deploy AI-powered DNS filtering for real-time malware blocking
- ✓ Implement DNSSEC for cryptographic authentication
- ✓ Monitor DNS queries with ML-based anomaly detection
- ✓ Maintain redundant, geographically distributed DNS infrastructure
- ✓ Conduct regular security audits and penetration tests
- ✓ Educate teams on DNS security risks and threats

**Secure DNS.
Secure Business.
Secure Future.**

Thank you

